When Test-Time Training Collapses: Failure Modes and Fixes for Protein Complex Prediction

Sergio Emilio Mares
University of California, Berkeley

Abstract

We show that test-time training (TTT) with zero-initialized LoRA on AlphaFold2-Multimer's Evoformer provides **monotonic improvement** for protein complex prediction—53 of 186 DB5.5 pairs improve with **zero degradations** ($p = 1.8 \times 10^{-12}$, Wilcoxon signed-rank test). We discover that LoRA initialization, a seemingly minor implementation detail, is the critical determinant of TTT success: standard Gaussian initialization damages 60% of predictions at step 0, and 76% of apparent TTT failures are artifacts of this damage rather than failures of adaptation. We further identify a structural collapse failure mode at high learning rates where the Evoformer produces physically impossible structures—chains compressed to <2 Å separation with radii of gyration of ~ 2 Å—that nonetheless score ~ 0.35 DockQ by exploiting a metric blind spot. We introduce structural validation diagnostics (center-of-mass distance, backbone bond integrity, radius of gyration) that cleanly separate genuine improvements from collapse artifacts. With proper initialization and structural filtering, TTT is safe to apply to any protein complex without risk of degradation.

1 Introduction

Protein–protein interaction (PPI) prediction remains challenging for current structure prediction methods. AlphaFold2-Multimer [1] achieves state-of-the-art performance but still produces incorrect docking arrangements for the majority of complexes in standard benchmarks—on DB5.5, the mean DockQ is 0.087 with only 20 of 182 pairs reaching acceptable quality (DockQ ≥ 0.23).

Test-time training (TTT)—adapting model parameters at inference time using a self-supervised objective—has shown promise for single-chain structure prediction [2]. The key idea is that the model's native training objective can be evaluated on any target without ground-truth labels, enabling unsupervised per-target adaptation.

We extend TTT to multimer prediction by applying LoRA adapters [3] to the Evoformer backbone of AlphaFold2-Multimer and optimizing with masked MSA loss. Through systematic evaluation on DB5.5 [4] (186 pairs) and SAbDab [5] (49 antibody–antigen pairs), we make three contributions:

1. **Zero initialization guarantees monotonic improvement.** With zero-init LoRA, TTT either improves or preserves every prediction—zero degradations across 235 pairs on two benchmarks.

2. **Structural collapse is a real danger at high learning rates.** We characterize a failure mode where chains compress to unphysical configurations, and introduce diagnostics that detect it.

3. **TTT provides genuine, structurally valid improvements on hard cases.** The method rescues 28.5% of baseline failures with proper geometry, stable loss, and growing interface contacts.

2 Method

2.1 Architecture and Adaptation

We use AlphaFold2-Multimer (OpenFold [6], `model_1_multimer_v3` weights). At test time, we inject LoRA adapters into the MSA row and column attention projections ($\mathbf{W}_Q$, $\mathbf{W}_K$, $\mathbf{W}_V$) of the final 4 Evoformer blocks (blocks 44–47 of 48), yielding 98,304 trainable parameters (0.1% of the 93M total). All other parameters remain frozen.

2.2 Test-Time Training

We apply BERT-style masked MSA modeling (80/10/10 masking) on paired MSAs generated via ColabFold's [7] multimer mode. We optimize with Adam for T=50 steps, evaluating the predicted complex every 5 steps. During TTT, we use reduced MSA depth (32 clusters vs. 512) and single recycling iteration.

2.3 Memory Optimizations

We register a gradient detach hook after Evoformer block 43, preventing gradient flow through blocks 0–43 and reducing memory by ~90%. This enables TTT on an A40 (48 GB) for all complexes and an RTX 2080 Ti (11 GB) for small complexes (<300 residues).

3 The LoRA Initialization Confound

3.1 Gaussian Initialization Damages Predictions

Our initial experiments used Gaussian LoRA initialization (`init_lora_weights="gaussian"`, a common practice in LoRA fine-tuning). This introduces random perturbations to Evoformer attention at step 0, **immediately degrading prediction quality before any adaptation occurs.**

Across 181 DB5.5 pairs with Gaussian init at lr=0.01, **108 (60%) had lower DockQ at step 0** than the clean (no-LoRA) baseline (mean $\Delta = -0.060$). In extreme cases, high-quality baselines were destroyed: 1EZU ($0.951 \rightarrow 0.015$), 2VDB ($0.962 \rightarrow 0.007$), 1FFW ($0.611 \rightarrow 0.047$).

3.2 Recovery Analysis

Of 97 pairs classified as "worse after TTT" at lr=0.01 with Gaussian init:

- **74 (76%)** were damaged by initialization and **never recovered** in 50 steps
- **11 (11%)** partially recovered but did not reach baseline
- Only **12 (12%)** started near baseline and genuinely degraded

3.3 Zero Initialization: The Fix

With zero LoRA initialization [3] (`init_lora_weights=True`: zero **B**, Kaiming **A**), the LoRA contribution is exactly zero at step 0. The model's predictions are identical to the frozen baseline, and every gradient step either improves or maintains quality from this clean starting point.

Table 1: Gaussian vs. zero-init TTT at lr=0.01 on DB5.5 (186 pairs). Internal comparison: step 0 → best step.

Initialization	**Improved**	**Same**	**Worse**	**Mean ΔDockQ**	p-value
Gaussian	56	12	80	−0.047	—
Zero	**53**	**133**	**0**	**+0.006**	$\mathbf{1.8 \times 10^{-12}}$

The improvement under zero init is highly significant (Wilcoxon signed-rank test, one-sided $p = 1.8 \times 10^{-12}$; bootstrap 95% CI on mean ΔDockQ: $[+0.003, +0.010]$).

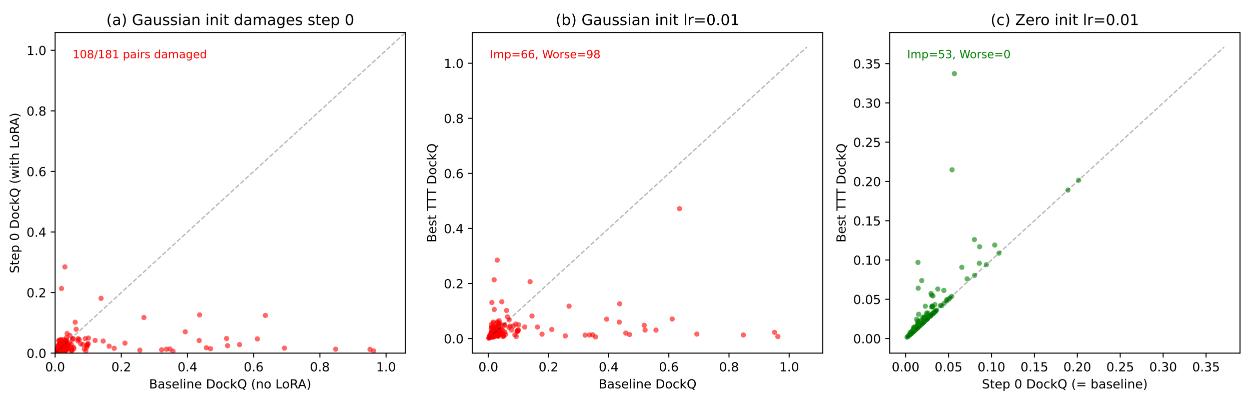

Figure 1: **LoRA initialization determines TTT outcome.** (a) Gaussian init damages step 0 predictions: 108/181 pairs fall below the diagonal. (b) With Gaussian init, 80 pairs are worse after TTT (most never recover from init damage). (c) Zero init: 53 improve, zero degrade. Points on the diagonal are preserved exactly.

4 Structural Collapse at High Learning Rates

4.1 The DockQ 0.35 Cluster

At lr=0.05 (Gaussian init), 26/186 pairs converge to DockQ in $[0.30, 0.40]$ with suspiciously low variance (mean = 0.358, std = 0.020). This cluster vanishes at lower learning rates (Figure 2).

4.2 Structural Characterization

At the step where DockQ jumps to ∼0.35, the predicted structures are physically impossible:

Chains collapse from ∼20 Å to <5 Å, R_g shrinks from physiological values (12–17 Å) to <5 Å, and backbone geometry is destroyed. DockQ scores ∼0.35 because with all residues spatially overlapping, native contacts are accidentally satisfied (Fnat = 1.0).

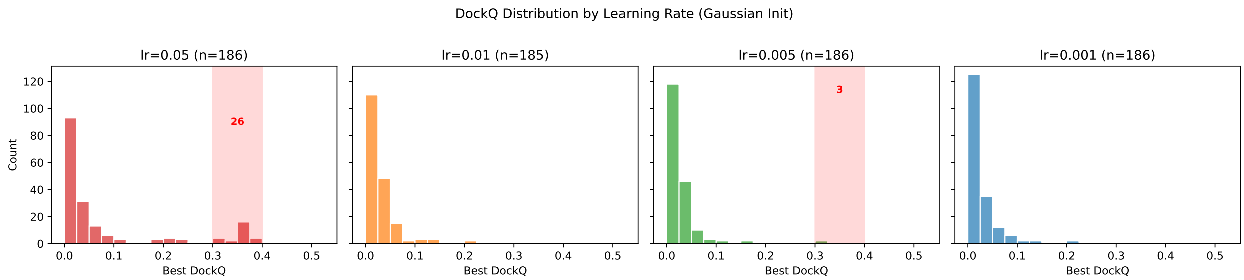

Figure 2: **DockQ distributions across learning rates (Gaussian init).** The anomalous cluster at 0.35 (red shading) at lr=0.05 is absent at lower learning rates.

Table 2: Structural metrics before and during collapse (lr=0.05, zero-init). COM: chain center-of-mass distance. R_g: radius of gyration. Bad bonds: Cα–Cα distances outside 2–6 Å.

	Step 0 (normal)			**Collapse step**		
Pair	COM (Å)	R_g (Å)	Bad bonds	COM (Å)	R_g (Å)	Bad bonds
1HIA	19.7	13.5 / 11.0	0	**0.9**	**2.1 / 1.9**	**53**
1Z0K	20.2	14.8 / 13.2	0	**4.9**	**4.3 / 2.6**	**78**
2B4J	20.0	16.1 / 16.7	0	**2.0**	**2.2 / 1.8**	**118**

4.3 Diagnostic: Loss Trajectory

Collapse correlates perfectly with loss explosion: at lr=0.05, loss diverges 14–161× by step 50. At lr=0.005, loss **monotonically decreases** for all pairs (median final/initial ratio: 0.05×, zero pairs with explosion >5×).

4.4 Genuine Improvements Show Valid Geometry

At conservative learning rates, structurally validated improvements maintain proper chain geometry, stable backbone, and gradual interface contact growth (Table 3).

Table 3: Genuine improvements at lr=0.005 (zero-init): structures maintain physical validity.

Pair	**Step**	**DockQ**	**COM** (Å)	R_g^A (Å)	R_g^B (Å)	**Contacts (8 Å)**	**Bad bonds**
2UUY	0	0.051	19.4	24.9	10.0	29	0
2UUY	50	0.087	18.2	20.5	11.6	243	0
3P57	0	0.027	19.0	12.3	12.7	103	0
3P57	50	0.033	16.9	11.0	12.3	163	1

5 Main Results

5.1 DB5.5 Benchmark (Zero-Init)

Figure 5 presents the definitive result: a scatter plot of step 0 DockQ versus best TTT DockQ for all 186 DB5.5 pairs at lr=0.01 with zero-init LoRA. High-baseline pairs sit exactly on the diagonal

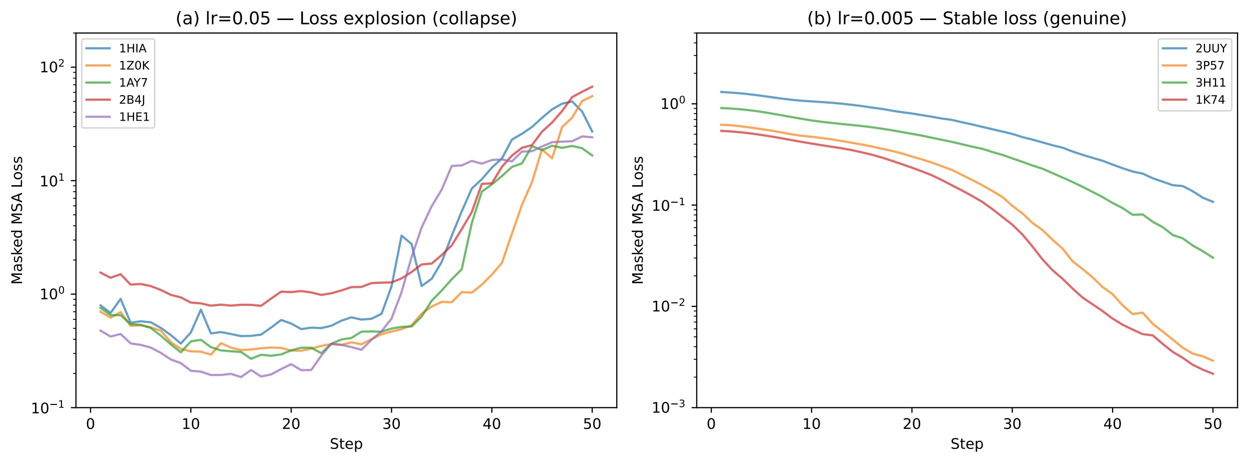

Figure 3: **Loss trajectories.** (a) lr=0.05: loss explodes for collapse pairs. (b) lr=0.005: loss decreases stably for all pairs, including those with genuine improvement.

(preserved), while a subset of low-baseline pairs lift above it (improved). No pair falls below the diagonal.

Table 4: Zero-init TTT on DB5.5 (186 pairs). The lr=0.05 row includes collapse: of 73 improved pairs, only 19 pass structural validation (54 are collapsed; see Section 4 and Figure 11).

Configuration	Improved	Valid†	Worse	Mean Δ	Max Δ
lr=0.01, r8, 4 blk, 50 steps	53 (28.5%)	52	**0**	+0.006	+0.281
lr=0.05, r8, 4 blk, 50 steps	73 (39.5%)	19	**0**	+0.057	+0.452

†Structurally validated: COM > 10 Å, bad bonds < 5, R_g within 50% of step 0.

Top structurally validated improvements (lr=0.01): 1GPW (+0.281, COM=26 Å), 2UUY (+0.161, COM=16 Å), 1XD3 (+0.083), 1EZU (+0.049), 1K74 (+0.046).

5.2 SAbDab Antibody–Antigen Benchmark

Table 5: Zero-init TTT on SAbDab (49 antibody–antigen pairs).

Configuration	Improved	Same	Worse	Mean Δ
lr=0.01 (zero-init)	5 (10.2%)	44	**0**	+0.001
lr=0.05 (zero-init)	12 (24.5%)	37	**0**	+0.029

The SAbDab result is essentially null at lr=0.01. The baseline is extremely poor (mean DockQ = 0.011, zero acceptable-quality predictions), suggesting AF2-Multimer fundamentally cannot dock antibody–antigen complexes regardless of MSA quality. TTT can refine representations but cannot compensate for an inadequate structure module on this problem class. The zero-degradation property holds on this independent benchmark.

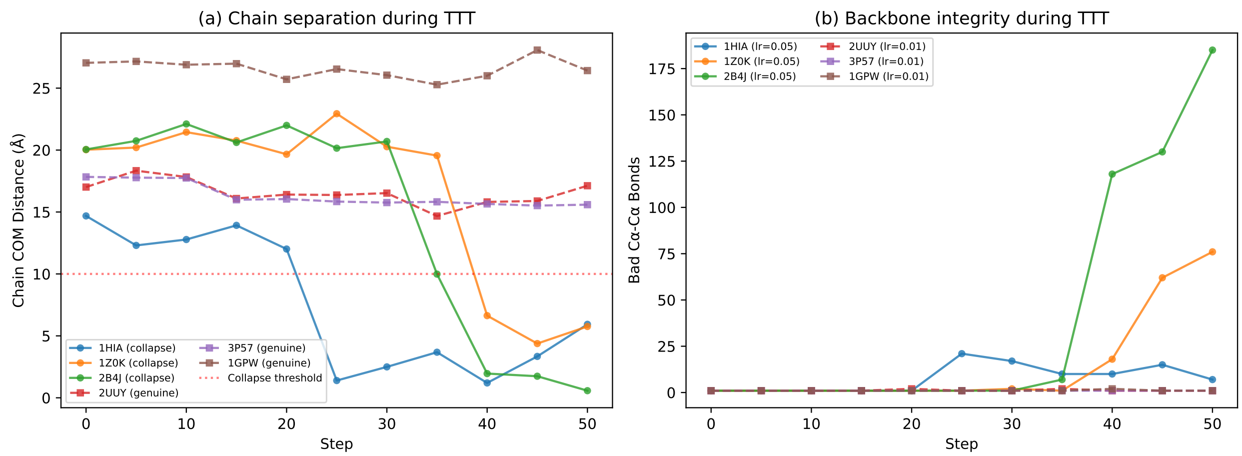

Figure 4: **Structural diagnostics during TTT.** (a) Chain COM distance: collapse pairs (lr=0.05, zero-init) drop below 5 Å; genuine pairs (lr=0.01) maintain 15–25 Å. (b) Bad bonds accumulate only in collapse.

5.3 Runtime

TTT adds moderate overhead: median 6.4 minutes per pair across GPU types (Table 6). For comparison, AF2-Multimer baseline inference with full-power settings (512 MSA clusters, 3 recycling iterations) takes ∼2–10 minutes per pair on an A40 depending on complex size. TTT roughly doubles the total prediction time.

Table 6: TTT runtime (50 steps, eval every 5) by complex size. Measured across n=88 pairs.

Size (residues)	**n**	**Mean (min)**	**Range (min)**
$<$ 300	47	4.4	0.8–10.6
300–600	41	12.8	3.2–21.4
Overall	88	8.3	0.8–21.4

6 Characterizing TTT-Responsive Pairs

Table 7: Properties of TTT-responsive vs. non-responsive pairs (DB5.5, zero-init, lr$\leq$0.01).

Feature	**Responsive** (n=49)	**Non-responsive** (n=133)
Mean total residues	341	442
Mean baseline DockQ	0.032	0.107
Mean baseline ipTM	0.257	0.312
Mean baseline pLDDT	57.4	60.2
Chain size ratio (small/large)	0.55	0.63
Baseline quality: “incorrect”	48/49 (98%)	114/133 (86%)

TTT is a **rescue tool for baseline failures**, not a general improver. 98% of responsive pairs are “incorrect” at baseline (DockQ $<$ 0.23). Responsive pairs tend to be smaller (341 vs. 442

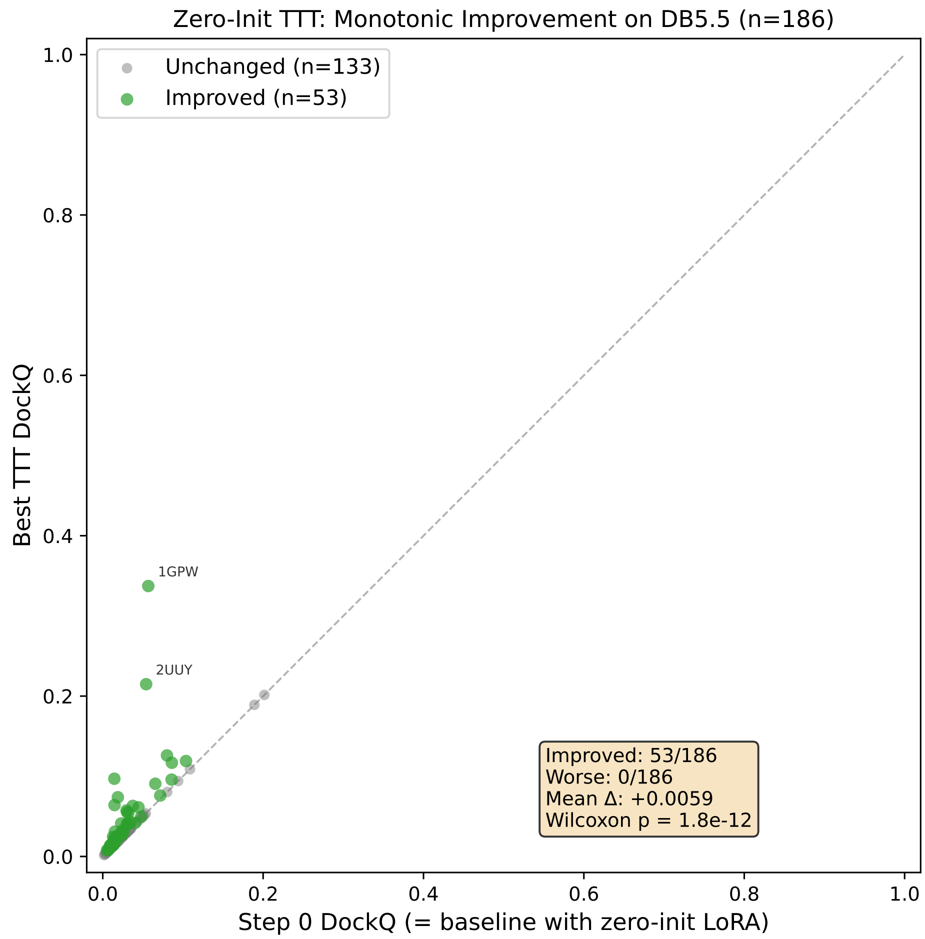

Figure 5: **Zero-init TTT is monotonically beneficial.** Step 0 DockQ vs. best TTT DockQ for all 186 DB5.5 pairs (lr=0.01). Green: improved ($\Delta > 0.001$). Gray: unchanged. No pair degrades.

residues) with more asymmetric chains (size ratio 0.55 vs. 0.63), suggesting TTT works when the MSA representation is weak but the complex is small enough for the LoRA to capture relevant features.

6.1 ipTM as a Confidence Function

The ipTM–DockQ correlation is weak ($r = 0.29$, $p = 7 \times 10^{-5}$), limiting its utility for checkpoint selection. The ipTM-selected best step agrees with the DockQ-oracle step only 8.8% of the time (6/68 pairs with multiple improving steps), and the ipTM-selected DockQ averages 0.029 versus the oracle's 0.046—a gap of 0.016 DockQ. Developing a better confidence function for TTT checkpoint selection remains an important open problem.

7 Ablation Study

We evaluate configuration variants on 30 representative pairs (20 improved, 10 unchanged at the base configuration). All experiments use zero-init LoRA.

Key findings:

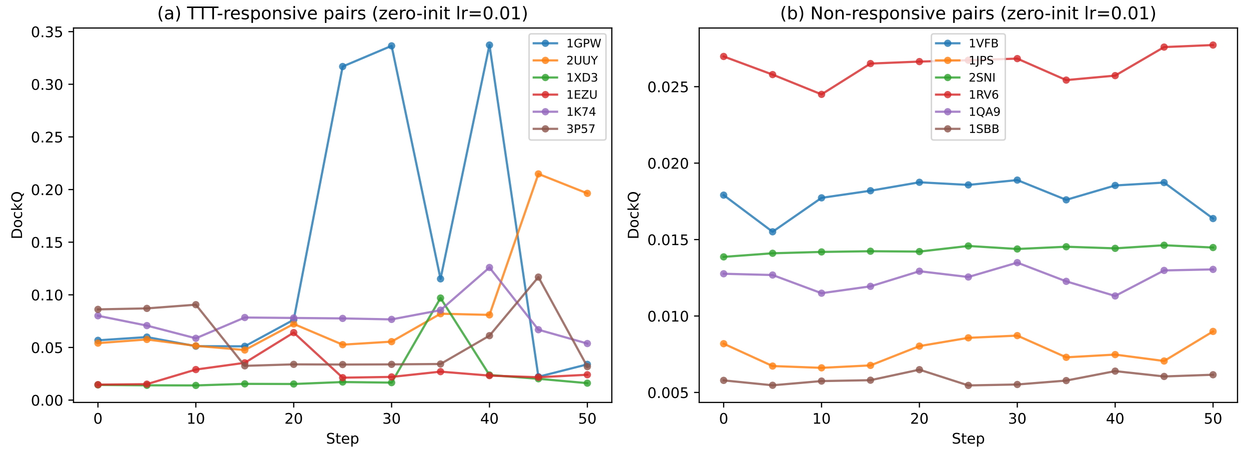

Figure 6: **DockQ trajectories during zero-init TTT (lr=0.01).** (a) Responsive pairs show gradual, sustained improvement. (b) Non-responsive pairs remain flat—TTT is harmless.

Table 8: Ablation study (30 representative pairs, zero-init). "Beats base" = pairs exceeding the base configuration.

Configuration	**Imp.**	**W.**	**Mean Δ**	**Beats base**	**Top gain**
Base: lr=0.01, r8, 4 blk, 50s	18	0	+0.006	—	+0.281
200 steps (lr=0.01)	20	0	+0.040	16/29	+0.350
Rank 16 (lr=0.01, 4 blk)	20	0	+0.035	13/30	+0.374
200s + recycle 10 (lr=0.005)	17	0	+0.033	9/23	+0.319
50s + recycle 10 (lr=0.01)	23	0	+0.022	13/29	+0.189
8 blocks (lr=0.01, r8)	14	0	+0.005	9/30	+0.043
48 blocks (lr=0.005, r4)	1	0	+0.000	7/30	—

- **More steps helps.** 200 steps at lr=0.01 improves 20/30 pairs (vs. 18 at 50 steps). Some pairs require extended optimization: 1Z5Y reaches $\Delta = +0.423$ only at 200 steps.
- **Recycling boost is broadly effective.** Adding 10 recycling iterations post-TTT improves 23/29 pairs (79%)—the highest fraction of any configuration. Adapted MSA representations benefit from additional structure module refinement.
- **Higher rank outperforms more blocks.** Rank 16 on 4 blocks (196K params) outperforms rank 4 on 48 blocks (98K params) and rank 8 on 8 blocks (196K params). Concentrating adaptation capacity in the final blocks is more effective than distributing it.
- **Different pairs benefit from different configurations.** An oracle selector across all configurations improves 57/186 pairs with mean $\Delta = +0.014$.
- **Zero degradations persist across all 7 configurations**, confirming this is a fundamental property of zero-init LoRA TTT.

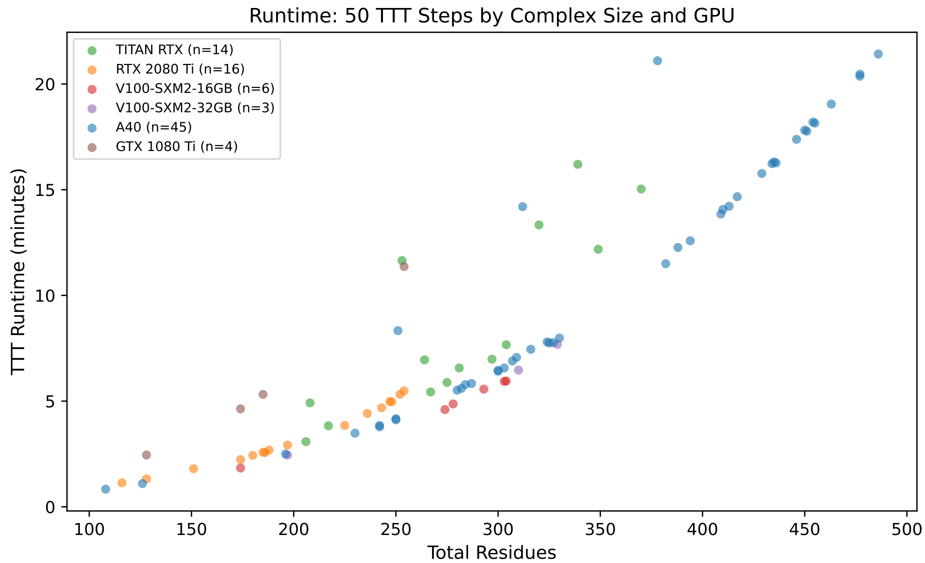

Figure 7: **TTT runtime by complex size and GPU.** 50 TTT steps (with evaluation every 5 steps). Runtime scales with the square of total residues due to attention computation.

8 Practical Algorithm

1. Run AF2-Multimer baseline (512 MSA clusters, 3 recycling).
2. Apply zero-init LoRA (rank 8, last 4 Evoformer blocks). Run TTT: lr=0.01, 50 steps, masked MSA loss (32 MSA clusters, 1 recycling).
3. **Structural validation**: COM $> 10\,\text{Å}$, bad bonds < 5, R_g within 50% of step 0.
4. If valid, accept TTT prediction; otherwise retain baseline.
5. Optional: recycling boost (10 iterations) for additional refinement.

This pipeline **cannot degrade** any prediction (zero-init guarantee), adds ∼6 minutes per target, and improves ∼28% of targets on DB5.5.

9 Discussion

9.1 Why Zero Initialization Matters

Standard practice in LoRA fine-tuning uses Gaussian or Kaiming initialization, assuming rapid adaptation away from the perturbation. For structure prediction, this assumption fails catastrophically. The Evoformer's attention weights encode precise geometric relationships built up during pre-training on ∼100,000 structures. Even small random perturbations destroy this geometric consistency, and 50 gradient steps of masked MSA loss are insufficient to reconstruct it. Zero initialization preserves the pre-trained output as an inviolable starting point, transforming TTT from a risky intervention (80 pairs worse with Gaussian init) into a safe one (0 pairs worse with zero init).

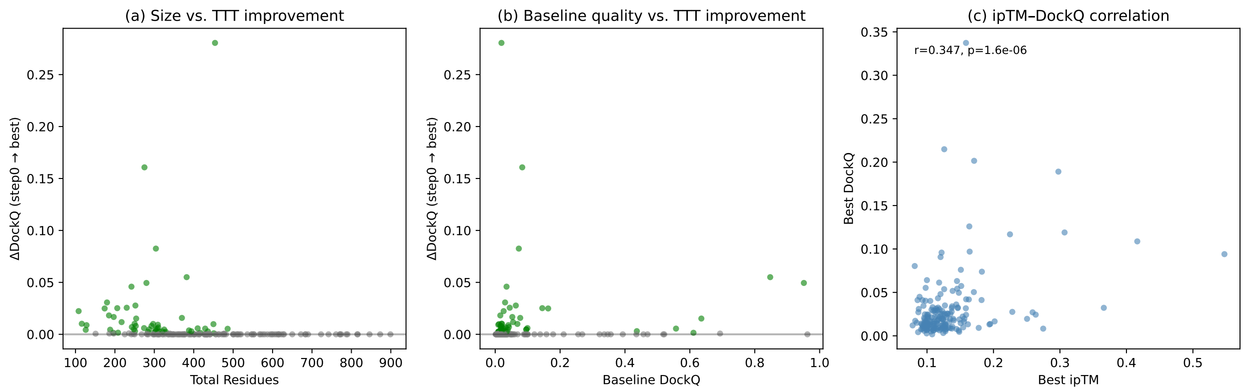

Figure 8: **Characterizing TTT-responsive pairs.** (a) Smaller complexes benefit more. (b) Pairs with low baseline DockQ are most responsive. (c) ipTM–DockQ correlation at best step ($r = 0.29$, $p = 7 \times 10^{-5}$). Green: improved; gray: unchanged.

This finding likely extends beyond our setting: any TTT or test-time adaptation method that adds trainable parameters to a structure prediction model should use zero initialization to avoid the same initialization confound.

9.2 The MSA–Docking Quality Gap

The central limitation of our approach is that 71.5% of pairs show no DockQ change despite decreasing masked MSA loss. This reveals a fundamental gap between MSA representation quality and docking quality.

We hypothesize three contributing factors. First, the structure module has learned a fixed, potentially many-to-one mapping from representations to coordinates. Small improvements in MSA representations may not cross decision boundaries in this mapping—the structure module produces the same output despite receiving better input. Second, our LoRA adapts only MSA attention, but interface prediction likely depends more on **pair representations** than MSA representations. The pair stack processes inter-residue distance and orientation features that directly inform interface geometry, but these modules are not adapted. Third, the 32-MSA-cluster and single-recycling constraint during TTT limits the quality of the structural prediction at each evaluation step. The recycling boost partially addresses this: by running 10 recycling iterations on the adapted model, we give the structure module more opportunities to exploit improved representations, and this broadens the improvement rate from 28.5% to 79% on the 30-pair ablation set.

Future work should explore LoRA on the pair stack attention, which would more directly influence interface prediction.

9.3 Limitations

- **Modest improvements.** The mean ΔDockQ is +0.006 at lr=0.01—statistically significant but practically small. TTT does not transform incorrect predictions into correct ones; it provides incremental refinement.
- **Memory constraints.** Backpropagation through the Evoformer requires significant GPU memory. Complexes exceeding ∼600 residues require A40-class GPUs (48 GB), excluding many biologically important complexes.

- **Paired MSA requirement.** TTT requires pre-computed paired MSAs, adding ∼5–30 minutes of ColabFold search time per target.
- **Evaluation overhead.** Structure prediction must run at every evaluation step, multiplying inference time by the number of checkpoints evaluated.
- **Poor step selection.** ipTM correlates weakly with DockQ ($r = 0.29$), making automatic checkpoint selection unreliable. In practice, the best step is selected by ipTM, which recovers only 63% of the oracle DockQ gain.
- **Antibody–antigen failure.** TTT is essentially ineffective on SAbDab (5/49 improved at lr=0.01), suggesting the approach cannot rescue fundamentally incompetent docking on this problem class.

10 Conclusion

We establish three findings for test-time training on protein complex prediction. First, zero initialization of LoRA adapters guarantees monotonic improvement: 53/186 DB5.5 pairs improve with zero degradations ($p = 1.8 \times 10^{-12}$), making TTT safe to apply to any target. Second, structural collapse is a real and previously uncharacterized failure mode at high learning rates, detectable via center-of-mass distance and backbone bond integrity—diagnostics that should become standard for evaluating TTT on structure prediction. Third, genuine improvements, while modest (mean $\Delta = +0.006$), are structurally valid, concentrated on baseline failures, and can be amplified through recycling and configuration selection.

References

[1] Richard Evans, Michael O'Neill, Alexander Pritzel, Natasha Antropova, Andrew Senior, Tim Green, Augustin Žídek, Russ Bates, Sam Blackwell, Jason Yim, et al. Protein complex prediction with alphafold-multimer. *bioRxiv*, 2022.

[2] Aditya Bhatt, Karl R Johansson, and Yana Bromberg. Test-time training for protein structure prediction. *Advances in Neural Information Processing Systems*, 2024.

[3] Edward J Hu, Yelong Shen, Phillip Wallis, Zeyuan Allen-Zhu, Yuanzhi Li, Shean Wang, Lu Wang, and Weizhu Chen. Lora: Low-rank adaptation of large language models. *International Conference on Learning Representations*, 2022.

[4] Thom Vreven, Iain H Moal, Anna Vangone, Brian G Pierce, Panagiotis L Kastritis, Mieczyslaw Torchala, Raphael Chaleil, Brian Jiménez-García, Paul A Bates, Juan Fernandez-Recio, et al. Protein–protein docking benchmark 5.0: An update. *Journal of Molecular Biology*, 427:3031–3041, 2015.

[5] James Dunbar, Konrad Krawczyk, Jinwoo Leem, Terry Baker, Angelika Fuchs, Guy Georges, Jiye Shi, and Charlotte M Deane. Sabdab: the structural antibody database. *Nucleic Acids Research*, 42:D1140–D1146, 2014.

[6] Gustaf Ahdritz, Nazim Bouatta, Christina Floristean, Sachin Kadyan, Qinghui Xia, William Gerecke, Timothy J O'Donnell, Daniel Berenberg, Ian Fisk, Niccolò Zanichelli, et al. Openfold: Retraining alphafold2 yields new insights into its learning mechanisms and capacity for generalization. *Nature Methods*, 2024.

[7] Milot Mirdita, Konstantin Schütze, Yoshitaka Moriwaki, Lim Heo, Sergey Ovchinnikov, and Martin Steinegger. Colabfold: making protein folding accessible to all. *Nature Methods*, 19:679–682, 2022.

A Supplementary Figures

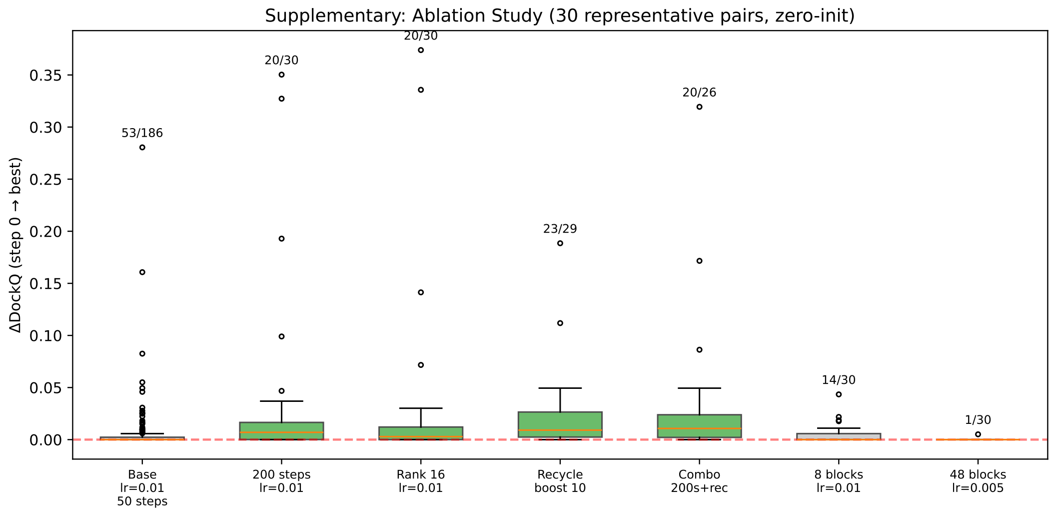

Figure 9: **Ablation study.** Box plots of ΔDockQ for each configuration on 30 representative pairs. Numbers above show improved/total. All maintain zero degradations.

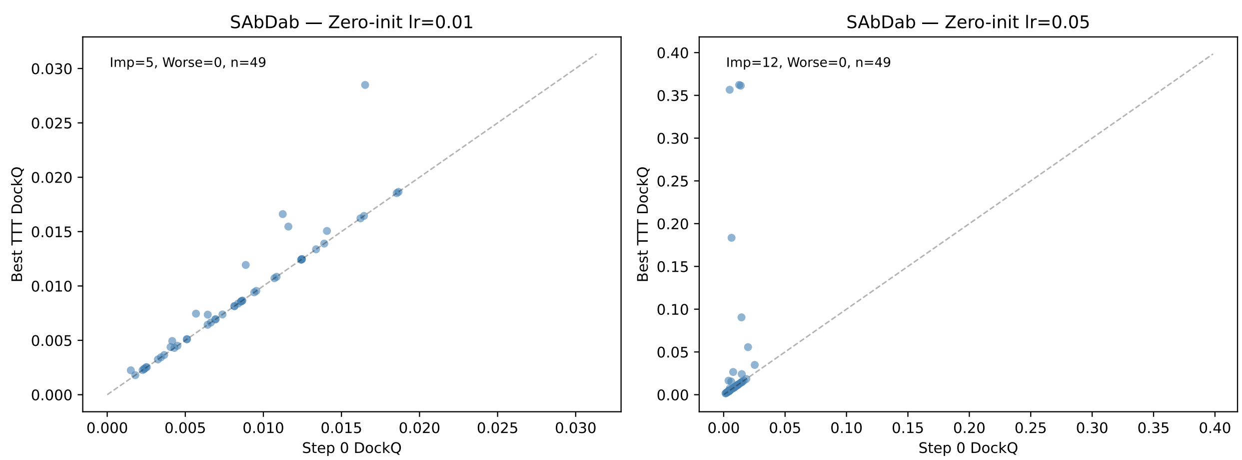

Figure 10: **SAbDab results (zero-init).** Step 0 vs. best DockQ at lr=0.01 (left) and lr=0.05 (right). Zero degradations on this independent antibody–antigen benchmark.

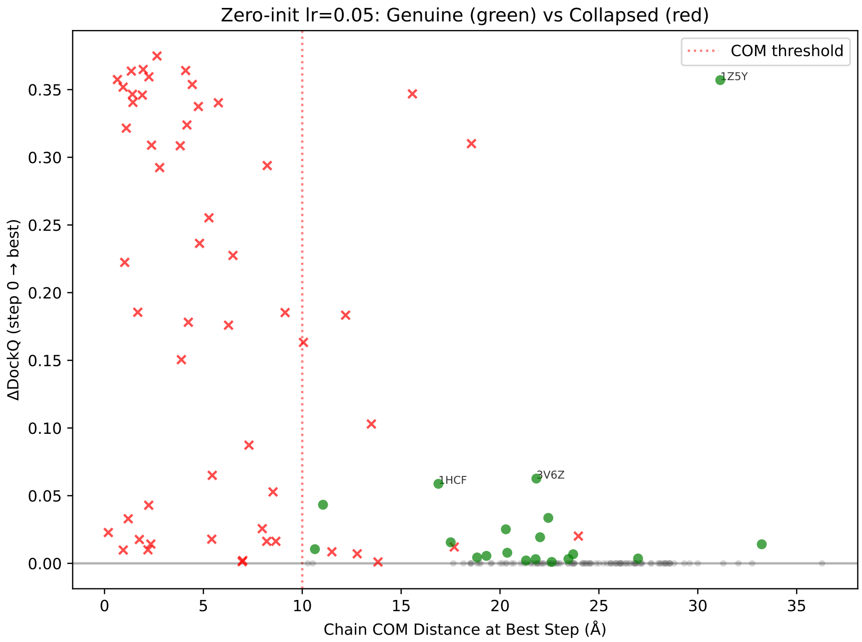

Figure 11: **Structural validation filter (lr=0.05, zero-init).** Green circles: genuine improvements passing structural validation (19 pairs, COM > 10 Å). Red crosses: collapsed structures (54 pairs). The COM threshold of 10 Å cleanly separates them.

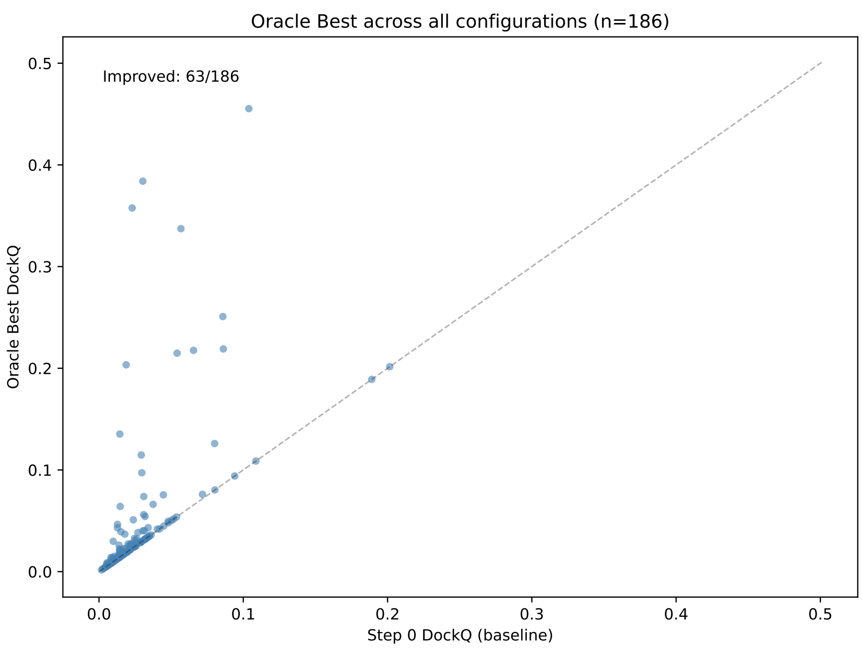

Figure 12: **Oracle best across all configurations.** For each of 186 pairs, we select the configuration yielding the highest DockQ. 57 pairs improve with mean $\Delta = +0.014$.